

Yipeng Ouyang

Undergraduate Researcher · New Energy Science and Engineering

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EDUCATION

Nanjing University

Sep. 2023 – Present

B.S. in New Energy Science and Engineering

Major courses: Materials Science, Optics, Thermology, Electrochemistry, Catalysis, Physical Chemistry.

Yale University — Summer Research Visiting Student

Jul. 2026 – Present

RESEARCH EXPERIENCE

Non-Precious-Metal-Based Proton Exchange Membrane Water Electrolysis for Hydrogen Production —

Undergraduate Researcher, Feb. 2025 – Present

Supervisors: Prof. Zhaosheng Li & Prof. Jianyong Feng (Group of Prof. Zhigang Zou)

- Engineered a novel cobalt-based composite catalyst integrating transition-metal oxides to overcome the intrinsic instability of non-precious metals in acidic OER environments.
- Investigated OER mechanism and material structure by conducting characterizations (TEM, SEM, XRD, XPS, XAS, DEMS and in-situ FTIR) and processed data into high-quality figures.
- Used rotating ring-disk electrode (RRDE) to study the catalyst surface-pH evolution.
- Fabricated high-performance catalyst-coated membranes (CCM) via optimized sol-gel synthesis and ink formulation for PEMWE integration; achieved >800 h stable operation at 100 mA cm⁻² with a 1.70 V cell voltage.
- Manuscript in preparation for submission (expected late 2026, as Joint First Author).

Yale University Summer Research — Scalable Perovskite-Si Photoelectrocatalysis for Water Splitting (DOE

EERE 3-year progress / Hu, Shu Lab / Kaidou Zhu Lab), Jul. 2026 – Present

- Designed perovskite-Si photoelectrocatalysis (PEC) and scanning electrocatalysis microscopy (SECM) cells via 3D printing.
- Coated Pt catalyst for AEMWE device by sputtering; handled Si-wafer pretreatment, edge-sealing, packaging and gold deposition.

INTERNSHIP EXPERIENCE

Sunwoda Energy Technology Co., Ltd. — R&D; Department Intern, Jul. 2025 – Aug. 2025

- Conducted electrochemical performance and safety evaluations of lithium-ion batteries, including overcharge protection and charge-discharge cycling tests.
- Assisted in preparation and organization of technical documentation for CNAS certification.

ADDITIONAL SKILLS

Technical: Athena, Fusion, Origin, Digital Micrograph, MDI Jade, Advantage, etc.

Languages: English (TOEFL iBT 90), Mandarin (Native)

Interests: Basketball (Guard), Electric guitar (Band member)

Research interests: Automation, Photoelectrocatalysis, Electrocatalysis